SAFE AND ROBUST REINFORCEMENT LEARNING IN TRANSPORTATION AGENTS DECISION-MAKING WITH REAL-TIME CONSTRAINT SATISFACTION.

By

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Introduction

Learning through interaction with the environment is the fundamental concept when considering the nature of intelligence, starting from the moment a primate or human infant is born. Young primates and human infants learn to interact through observation and feedback: Baby monkeys use exaggerated lip-smacking, neonatal imitation, and sustained mutual gaze, while human infants cry, play and wave. In these instances, there is no explicit teacher. Instead, they establish a direct sensorimotor connection to their environment, gaining a wealth of information about actions, reactions, cause and effect, consequences, and the necessary steps to achieve a goal.

Extending this principle to adult life, whether its learning to hold a nuanced conversation, land a rocket booster safely on a pad, or drive a car, every new interaction and experience serves as a major source of knowledge about our surroundings and, ultimately our individual capabilities. This reliance on learning from interactions and experience forms the central, foundational idea of most modern theories of intelligence.

My research and development will focus on the Proximal Policy Optimization (PPO) algorithm. Among the multiple variants of Reinforcement Learning (RL) algorithms available, PPO was selected for its high simplicity, stability and its dual nature to support both discrete and continuous action spaces natively. This makes it an ideal framework for training complex transportation agents.

Background and Context

Transportation agents represent the next significant phase of Physical Artificial Intelligence (PAI). They pose one of the hardest problems to solve due to their direct involvement with human lives and the real world. My research, development and agent training will therefore focus on transportation industry, specifically on landing and driving application of the algorithm.

Proximal Policy Optimization (PPO) is a state-of-the-art algorithm in Reinforcement learning (RL). PPO and it’s equivalent RL algorithms are widely used across multiple industries, including finance, chemistry, biomedical, traffic control, gaming, robotics, energy, marketing, advertising, network engineering and cybersecurity. Importantly, PPO is also the core RL algorithm that currently enables frontier Large Language Models (LLMs) like OpenAI ChatGPT, Anthropic Claude and Google Gemini through techniques like Reinforcement Learning from Human Feedback (RLHF)

PPO is also an on-policy algorithm, which means the agent learns from its own experience. In this scenario, the transport agent follows its current policy to take an action and collects the resulting experience. This experience is then used to evaluate the policy value which in turn iteratively improves the subsequent policy. This entire method is implemented utilizing an actor-critic network.

Problem Statement

The core problem addressed by this research is the practical training of robust transportation agents for safety-critical tasks.

In addition to developing a comprehensive theoretical understanding of Reinforcement Learning (RL), the primary objective is to train multiple transportation agents using Proximal Policy Optimization

(PPO) across diverse simulated environments. These environments such as OpenAI Gymnasium and Carla, feature integrated physics and world conditions designed to replicate real-world challenges. The successful training requires the agents to demonstrate the necessary intelligence to land and drive safely and successfully.

To ensure successful project execution, alignment and evaluation, the research will utilize the **SMART** framework to define project milestones and outcomes. This framework facilitates a clear and mutual understanding of what constitutes a satisfactory result. **SMART** stands for:

- **Specific:** What I intend to accomplish is clearly defined and focused.
- **Measurable:** There is quantifiable data to measure and most importantly, a criterion for success.
- **Achievable:** The project is realistic, within my skillset and available resources in Tongji University.
- **Relevant:** Why the project is important to me, its alignment with my skillset, career and experience.
- **Timebound:** Timeframe for the completion of the thesis and project implementation is realistic.

Finally, this work operates under the foundational principle that while RL algorithms continually evolve to improve efficiency and compensate for flaws, the underlying core fundamentals of learning from interaction and experience remain consistent.

Research Questions

This research and development project is designed to answer four major questions essential to the development of safe and effective transportation agents:

1. **System Design and Navigation:** How can a Proximal Policy Optimization (PPO) framework be effectively architected and trained to endow multiple transportation agents with the decision-making ability required to navigate safely across diverse simulated environments and terrains?

2. **Hyper-parameter and Reward Optimization:** What specific hyper-parameters and reward structures are necessary to guarantee that the trained transport agents operate with the maximum safety and functional efficiency within the designated simulated environments?

3. **Policy Evaluation and Success Criteria:** What objective and quantifiable metrics define a "well learned policy" for transportation agents in this context, and what level of performance is required to deem the training process successful?

4. **Reproducibility and Robustness:** Given the identical algorithmic code, hyper-parameters, and simulated environment, can a replicated training run consistently reproduce similar or demonstrably superior performance results?

Literature Review

I would be explaining the high-level concept of reinforcement learning, compare my implementation versus other researchers' vision and architecture and finally it's advantages and limitation.

Foundational Concepts and Theories

The Reward Hypothesis and the RL Framework:

The theoretical foundation of Reinforcement Learning is the Reward Hypothesis, which posit that "any goal can be formalized as the outcome of maximizing the cumulative reward." This optimization problem is solved through repeated interaction between an agent and its environment. This relationship is defined by the Agent-Environment Interaction loop

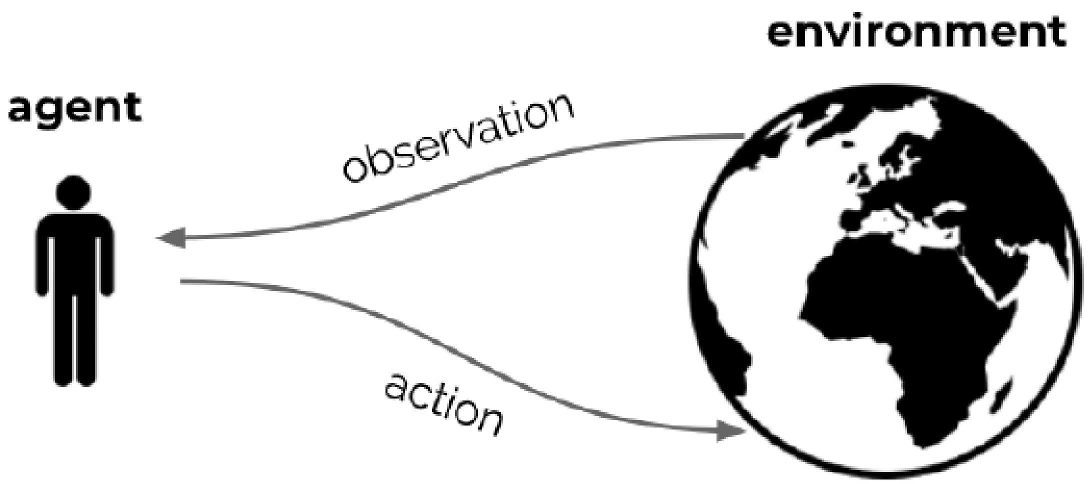


Figure 1: The Agent-Environment Interaction loop.

The process adheres to the logic of a sequential decision-making process. At each discrete time step t, the interaction loop proceeds as follows:

- **The Agent:**
  - Receives an observation O_t (representing the current state S_t) and a scaler reward R_t from the environment.
  - Based on its current policy, the agent selects and executes an action A_t.

- **The Environment:**
  - Receives the agent's action A_t.
  - Transitions to a new state, emitting a new observation O_{t+1} (or S_{t+1}).
  - Provides a corresponding immediate reward R_{t+1}.

The agent's task is thus to learn the optimal mapping from observations (O_t) to actions (A_t) that maximizes the long-term return (G_t).

Dynamic Programming (DP):

The underlying optimization problem of sequential decision-making is rooted in Dynamic Programming (DP). DP is a powerful optimization technique that solves complex problems by breaking them down into simpler, overlapping subproblems, relying on the principles of Optimality. DP methods such as Value iteration and Policy iteration are foundational to Reinforcement Learning theory; however, they require a perfect model of the environment. Modern RL algorithms, by contrast are Model-free, learning the optimal policy through trial-and-error sampling when the environment dynamics are unknown or too complex to model explicitly.

Markov Decision Process (MDP):

The sequential decision-making problem faced by the transportation agent is formally modelled as a Markov Decision Process (MDP). An MDP is defined by the tuple M= (S, A, P, R, Y).

- **S:** The set of all possible states (Observations Ot).
- **A:** The set of all possible actions A_t the agent can take.
- **P:** The transition probability function, P(s'|s,a), which governs the probability of transitioning to state s' after taking action a in state s.
- **R:** The reward function, R(s,a), which defines the immediate reward received.
- **Y:** The discount factor (where 0<=Y<=1), which determines the present value of future rewards.

The key value of the MDP is the Markov property, which states that the future state depends only on the current state and action, not on the entire history of preceding events. This simplification makes the problem computationally tractable.

Proximal Policy Optimization (PPO)

To solve the MDP, the proposed research utilizes the leading model-free Actor-Critic algorithm known for providing the best balance between implementation simplicity, fast convergence and sample efficiency.

- ***Actor-Critic Architecture***: PPO utilizes two neural networks:
    - The Actor (π)suggests the best action to take given the current state.
    - The Critic (V) estimates the value of the current state, which is used to guide the Actor's update.
- ***Trust Region***: PPO is an improvement over Trust Region Policy Optimization (TRPO). It prevents destructive policy updates by enforcing a proximal constraint on the change between the new and old policy. This is achieved by using a clipped objective function, which ensures the new policy does not stray too far from the old one, thus maximizing stability and ensuring reliable learning.

Comparative Analysis and Implementation Debates

Implementing Reinforcement Learning for complex real-world tasks involves a very key choice regarding the level of abstraction. The following analysis compares three primary implementation strategies, justifying the custom approach selected for this research and development.

Level 1: High-Level Abstraction (Stable Baseline3)

The first approach utilizes highly abstracted, community-driven libraries, such as Stable Baseline3. These libraries are popular because they abstract complex mathematics and engineering, allowing for fast prototyping, benchmarking and execution of experiments with minimal code. Key advantages include:

- Ease of Use and Reliability: Facilitates fast prototyping and offers reliable, benchmarked implementations.
- Built-in Features: Includes native support for vectorised environments and TensorBoard logging.

However, disadvantages include limited customization, significant abstraction overhead, and the framework being tied exclusively to the pytorch library, which may restrict deployment options.

Level 2: Custom Algorithm Implementation (Intermediate)

The next level involves deeper implementation using native framework like Pytorch, TensorFlow or Jax. This involves using highly transparent, single-file implementation (Such as CleanRL) or writing the core algorithmic code and using it to train a single policy for different agents simultaneously. This is done by launching multiple non-interacting instances of an environment within the larger vectorized Gymnasium environments (e.g. Boxing, HalfCheeter, Ant).

This approach offers a better level of customization that level 1, but still limits agents' autonomy. Since all agents in the vector share a single set of weights, they are all tied to a single general policy, offering less flexibility and control than the level 3 custom approach.

Level 3: Full Custom Production-Ready Pipeline (Selected Approach)

The chosen strategy for this research and development is to build a full custom Reinforcement Learning pipeline suitable for production and tailored to the unique demands of safe transportation. This extensive pipeline encompasses all components, including but not limited to:

- **Core Architecture**: Agent, Policy-Value Network Architecture, and policy optimization.
- **Data Management**: Environment vectorization, State and Action Space definition, Data Generation, Data Processing and Storage, and the use of Generalized Advantage Estimate (GAE).
- **Evaluation and Deployment**: Implemented entirely using the TensorFlow framework.

Justification: This level of customization offers full control and maximum flexibility over every aspect of the algorithm, facilitating a deeper theoretical understanding, and enabling tailored performance for safety critical systems. The inherent trade-offs, however, include the high developmental cost in terms of time, increased potential for errors and bugs in new code, and the lack of direct public baseline since the system is built from the ground up.

Gaps in the Robust RL for Transportation Research

While PPO is highly effective, its direct application to safety-critical transportation agents introduces specific challenges that defines this research gap:

1 **Safety as a Constraint, not just a reward:** Standard RL optimizes for cumulative reward, which may incentivise dangerous behaviour if a high reward is achievable only through risky actions. In transportation safety must be treated as a hard, real-time constraint that must never be violated.

2 **Lack of Real-Time constraint satisfaction:** Current PPO implementations often lack a robust, integrated mechanism for evaluating and ensuring the satisfaction of dynamic, real-world constraints (e.g. speed limits, collision avoidance distance) within the agent's decision loop.

3 **The Sim-to-Real Gap:** While the actual RL skills and techniques are transferrable to the real world, a significant gap exists in transferring policies learned in digital twins, controlled environments or simulators to the unpredictable real-world scenarios. The fidelity mismatch between these environments and the real-world test results in policies that are brittle and fail to generalize outside the training environment.

4 **Reproducibility and Robustness:** The complexity of large-scale simulators demands that the learned policy must be highly robust to the environment noise and reproducible across multiple training runs, a common challenge in cutting-edge RL research.

Bridging the Gap (Workarounds)

The research would address the identified gaps using the following methodological solutions:

1 **Constraint Policy Optimization (CPO)/ Constraint Integration:** For gap 1, the optimization problem would be modified from simple reward maximization to maximizing reward subject to a constrain on risk (e.g. maximum allowed constraint violation or risk tolerance).

2 **Augmented Reward/Loss Function:** For gap 2, a penalty term will be explicitly integrated into the PPO loss function. This term will penalize the agent whenever a specified real-time constraint (e.g. minimum stopping distance) is violated, forcing the policy to prioritize compliance.

3 **Domian Randomization/Domain Adaptation:** For gap 3, the simulator parameters (e.g. friction, sensor noise, lightening) will be randomly varied during training (Domain Randomization) to prepare the agent for the real-world. This forces the policy to learn features that are robust to uncertainty, improving its ability to generalize to the real-world.

4 **Extensive Hyperparameter Tuning and Seed Management:** For gap 4, policy stability would be maximized through rigorous hyper-parameter search (Grid Search/Bayesian Optimization) and meticulous management of random seeds across all training runs to ensure the learned policies are consistently reproducible.

Research Design and Methodology

The overall approach to this research and development is designed to facilitate the creation and robust, safety-aware transportation agents.

Research design: Three tiers of Implementation

The research design is strategically structured into three distinct tiers to ensure comprehensive development, rigorous experimentation and full control over the RL pipeline:

- **Custom Model Development**: This foundational tier involves the complete implementation and integration of the entire Proximal Policy Optimization (PPO) pipeline. This involves defining the compute environment, selecting the Operating System (Linux), setting up the development infrastructure, designing the Model architecture (Actor-Critic Network), and ensuring its compatibility with the specific AI use-case (transportation decision-making).
- **Simulation Development for Agent Training**: This tier utilizes established open-source simulation environments, specifically OpenAI Gymnasium and Carla, as the application layer. These platforms would be used to integrate the custom PPO algorithm and train the Reinforcement Learning agent to achieve the necessary navigational and operational intelligence under real-world-like physics and conditions.
- **Experimental Science and Policy Profiling:** This tier focuses on rigorous experimentation to maximize policy robustness and safety. It involves manipulating key variables, including training hyper-parameters, network size and custom reward/loss function.

Methods and Data Sources

The choice of PPO- an on-policy and Model-free Reinforcement Learning technique- dictates the data generation method. The necessary training data is not pre-collected; instead, it is generated, collected and continuously improved by the learning agent itself through its direct interaction with the simulated environments.

Tools, Budget and Practical Obstacles

The primary obstacle identified was the lack of readily available High-Performance Computing (HPC) resources within the department, whether via an internal datacenter or immediate access to a Cloud Service Provider (CSP) like Alibaba Cloud or Google Cloud. This issue has been resolved by purchasing and integrating a personal server with dedicated GPU capacity. This provides a stable, locally controlled compute resource where all the necessary open-source tools for the RL implementation and simulation training have been successfully integrated.

The self-sufficient laboratory combined with the project's SMART framework and fully customized pipeline, enables the research and development to proceed with minimal to no supervision, requiring only administrative oversight, assistance and support.

Financial Justification and Budget Impact

The decision to build a custom dedicated infrastructure offers significant financial advantages, demonstrating fiscal responsibility to the university.

Comparative Budget Analysis:

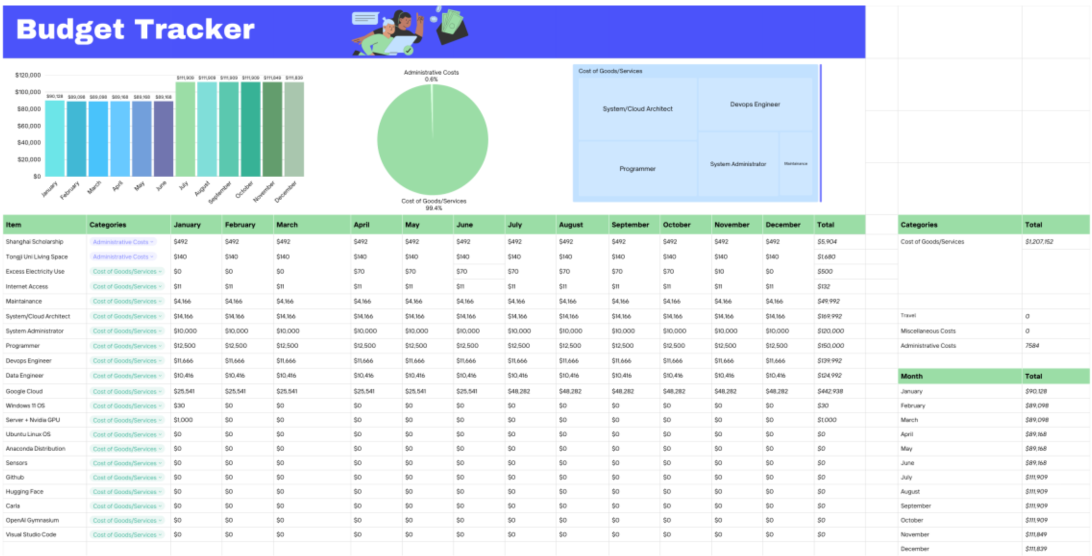
Alternative 1 (Cloud Compute| $1.7M – $2.1M) | 3-year estimate: This is for 1-3 researchers using major Cloud Service Provider (Google Cloud: 2,128,500 including SME fees; Alibaba Cloud: Approximately 1.7M). This is the cost I have averted by using a dedicated server.

Alternative 2 (Datacenter CapEx) |$6M – $100M estimate: Capital expenditure for building a high-performance datacenter, excluding the substantial annual Operational Expenditure (OpEx) for maintenance, power and engineers.

My solution (Dedicated Personal Server): $30k – $40k is the Total Actual Cost to Tongji University/ Shanghai Government for the doctorate program length.

This proposal represents a net cost reduction of over $1.67M for the Transportation Department/Tongji University when compared to the minimum cloud compute alternative. Furthermore, the use of this custom setup and Doctorate education in China avoids significant personal costs (estimated at $90k- $150k for a 3-year doctorate degree in Canada/United States) that would have been incurred.

The figure below shows the open-source tools and one year budget estimate of SME services



Item	Categories	January	February	March	April	May	June	July	August	September	October	November	December	Total
Shanghai Scholarship	Administrative Costs	$492	$492	$492	$492	$492	$492	$492	$492	$492	$492	$492	$492	$5,904
Tongji Uni Living Space	Administrative Costs	$140	$140	$140	$140	$140	$140	$140	$140	$140	$140	$140	$140	$1,680
Excess Electricity Use	Cost of Goods/Services	$0	$0	$0	$70	$70	$70	$70	$70	$70	$70	$10	$0	$500
Internet Access	Cost of Goods/Services	$11	$11	$11	$11	$11	$11	$11	$11	$11	$11	$11	$11	$132
Maintainance	Cost of Goods/Services	$4,166	$4,166	$4,166	$4,166	$4,166	$4,166	$4.166	$4,166	$4,166	$4,166	$4.166	$4,166	$49,992
System/Cloud Architect	Cost of Goods/Services	$14,166	$14,166	$14,166	$14,166	$14,166	$14,166	$14,166	$14,166	$14,166	$14,166	$14,166	$14,166	$169,992
System Administrator	Cost of Goods/Services	$10,000	$10,000	$10,000	$10,000	$10,000	$10,000	$10,000	$10,000	$10,000	$10,000	$10,000	$10,000	$120,000
Programmer	Cost of Goods/Services	$12,500	$12,500	$12,500	$12,500	$12,500	$12,500	$12,500	$12,500	$12,500	$12,500	$12,500	$12,500	$150,000
Devops Engineer	Cost of Goods/Services	$11,666	$11,666	$11,666	$11,666	$11,666	$11,666	$11,666	$11,666	$11,666	$11,666	$11,666	$11,666	$139,992
Data Engineer	Cost of Goods/Services	$10,416	$10,416	$10,416	$10,416	$10,416	$10,416	$10,416	$10,416	$10,416	$10,416	$10,416	$10,416	$124,992
Google Cloud	Cost of Goods/Services	$25,541	$25,541	$25,541	$25,541	$25,541	$25,541	$48,282	$48,282	$48,282	$48,282	$48,282	$48,282	$442,938
Windows 11 OS	Cost of Goods/Services	$30	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$30
Server + Nvidia GPU	Cost of Goods/Services	$1,000	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$1,000
Ubuntu Linux OS	Cost of Goods/Services	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Anaconda Distribution	Cost of Goods/Services	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Sensors	Cost of Goods/Services	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Github	Cost of Goods/Services	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Hugging Face	Cost of Goods/Services	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Carla	Cost of Goods/Services	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
OpenAI Gymnasium	Cost of Goods/Services	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0
Visual Studio Code	Cost of Goods/Services	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0

Categories	Total
Cost of Goods/Services	$1,207,152
Travel	0
Miscellaneous Costs	0
Administrative Costs	7584

Month	Total
January	$90,128
February	$89,098
March	$89,098
April	$89,168
May	$89,168
June	$89,168
July	$111,909
August	$111,909
September	$111,909
October	$111,909
November	$111,849
December	$111,839

Figure 2: Tools and 1 year Budget cost

Practical Implications and contributions to knowledge

Academic and Practical Implications:

This theoretical and practical research is significant because it addresses a critical gap in the field of Safe Reinforcement Learning (RL).

The primary contributions to knowledge would be:

- **Novel PPO Constraint Integration:** Developing and validating a novel, custom Constrained Policy Optimization (CPO) framework that ensures real-time safety constraints are met during autonomous decision-making.
- **Robust Sim-to-Real Methodology:** Providing a comprehensive methodology, utilizing Domain Randomization, for training highly robust transportation policies that are demonstrable reliable when faced with Sim-to-real-world fidelity mismatch.
- **Reproducible Safety Benchmarks:** Establishing high fidelity, reproducible benchmarks for the safe operation of transportation agents withing complex environments like OpenAI Gymnasium and Carla will aid future researchers.

Practical and Personal Implication:

Reinforcement Learning has broad practical applications, including a direct application to my core expertise in Network Engineering and Cybersecurity. Having spent over a decade in the industry, including roles at global firms such as Deloitte (Cybersecurity Consultant) and Cisco (Cybersecurity Systems Engineer), I recognised the critical need for reskilling and specialization in Artificial Intelligence to drive innovation and maintain a professional edge.

This research and development in RL is personally transformative. It not only provides specialized knowledge in the cutting-edge Autonomous Transportation, but also creates a synergistic “3 in 1” skill advancement, powerfully combining my experience in Network Engineering, Cybersecurity and now Physical Artificial Intelligence.

Educational Background

My educational background which supports the interdisciplinary nature of this research, is detailed as follows:

Doctor of Engineering (In view): Transportation Engineering, Tongji University, China
Master of Engineering (2016): Network Engineering, Toronto Metropolitan University, Canada
Bachelor of Science (2013): Computer Science, Babcock University, Nigeria

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[10] My open-sourced PPO training dataset: https://huggingface.co/privateboss

Research Development Roadmap

The project timeline is strategically structured to demonstrate feasibility and adherence to the Timebound element of the SMART framework and will proceed through a sequential, escalating path of complexity, focused on maximizing agent development and efficiency.

Phase 1

This phase demonstrates the essential foundational skills in simulation environment setup, sensor integration and complete Reinforcement Learning agent algorithm development and training required for the doctoral research.

Development Title	Objectives	Deadline
Modular Software-Defined Vehicle	Built an end-to-end self-driving car simulation in Carla for day/night scenarios focusing on topological and array-based sensor visualization.	**Completed.**
Lunar Lander (Discrete action spaces)	Applied the PPO pipeline to Train a Lunar Lander agent to land safely using Discrete action spaces in Gymnasium environment.	**Completed.**
Car-Race (Continuous RL)	Applied the PPO pipeline to train a single continuous policy car-racing agent to get the maximum score within the time constraint in Gymnasium environment.	**Completed.**

Phase 2

This phase details the development and training of agents required to satisfy the primary research questions, culminating in the completion of the project’s core development phase.

Development Title	Objectives	Deadline
Taxi (Discrete Control)	Implement the custom PPO pipeline in Gymnasium for Taxi to validate discrete constraint satisfaction and core logic stability	**January 30th 2026**
Lunar Lander (Continuous action spaces)	Implement the custom PPO pipeline to the Lunar Lander for Continuous action space (realistic physics/continuous action spaces focusing on advanced control	**April 30th 2026**
Agent 5	TBD	**September 30th 2026**
Agent 6	TBD	**TBD**